Androgen Induced Changes to Vocal Neurons of Female *Xenopus Laevis* Frogs: Preliminary Methodology

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Poster #

Introduction

- ***Xenopus laevis*** frogs produce sex dimorphic vocal behaviors, with males producing faster vocal pulses than females.
- Chronic testosterone treatment results in masculinization of the female vocal circuit[3], including enlargement of cells in PBx[2/4], a part of the central pattern generator responsible for *X. laevis* vocal calls.
- It is unsure how fast these changes occur, and whether increased T-treatment time correlates with larger soma size or other changes.

Goals

- ***Develop a timeline of anatomical changes to PBx in response to testosterone treatment***
- ***Develop an immunohistochemistry (IHC) protocol to discover possible testosterone induced neurogenesis in PBx***

Methods

Anatomy (24 Frogs)

Surgery
Ovariectomy followed by testosterone implantation into experimental frogs

Dissection
Frogs were sacrificed at a time point after surgery and then dissected to extract and examine the brains

Fluorescent Injection
The motor nucleus IX.X was injected with fluorescent dextran dye

Preparing Brains
Brains were fixed in formaldehyde and then cleared in SDS detergent

Brain Imaging
Done using light sheet microscopy

Neurogenesis (6 Frogs)

Surgery
Ovariectomy followed by testosterone implantation into experimental frogs

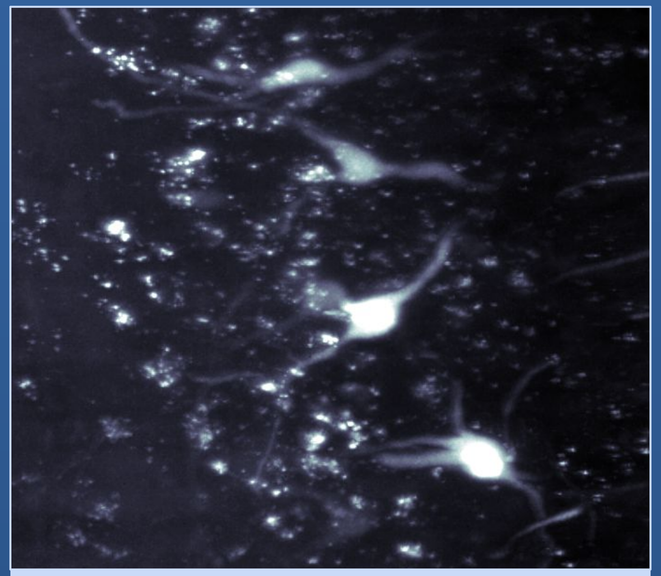
BrdU Administration
BrdU was injected into live frogs 2 days and 8 days after surgery

Dissection
Brains were extracted 2 weeks after surgery

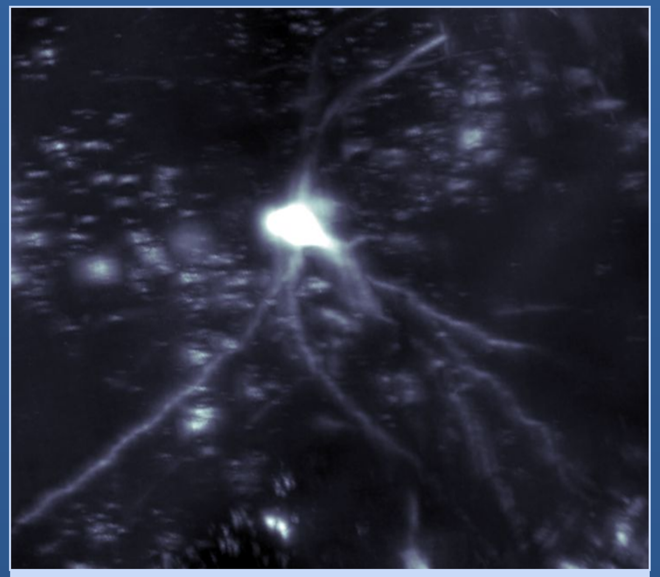
Preparing Brains
Brains were fixed in formaldehyde and then sectioned using a cryostat

BrdU Immunohistochemistry
Sections underwent an IHC protocol to stain BrdU incorporated cells

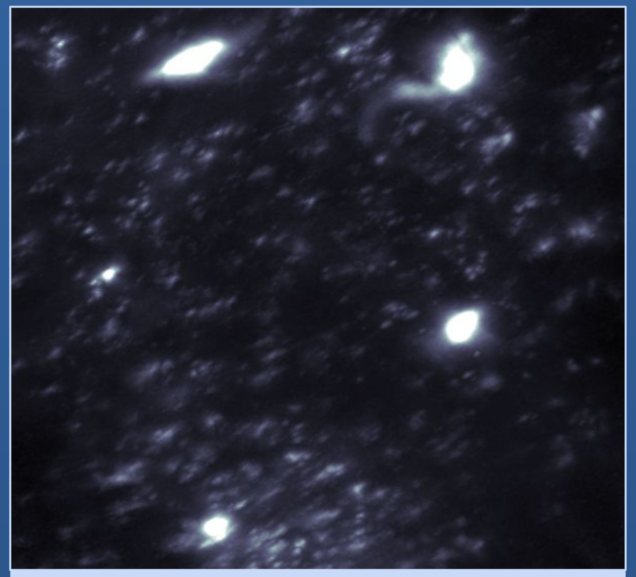
Brain Analysis
Brains were inspected for levels of neurogenesis in PBx



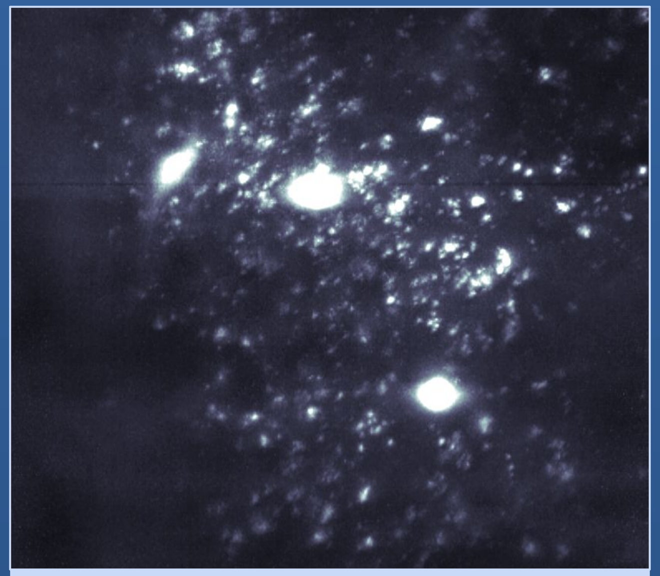
*PBx neurons in **control** group*



*PBx neuron in **two week** group*



*PBx neurons in **four week** group*

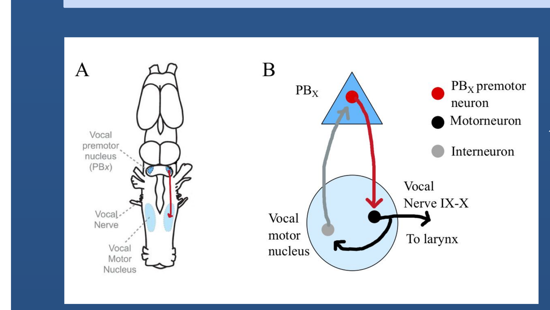


*PBx neurons in **six week** group*

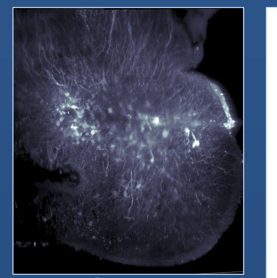
Imaging

After injection of micro-ruby 3000 mw fluorescent dye into n.IX.X, tissue was fixed and then cleared in 8% SDS at 37 C until visually transparent, and then put in a matching RI glycerol solution.

After a variable amount of time, brains were mounted and imaged using light sheet microscopy. Images were then analyzed using filament tracer function in Imaris.



Basic map of *X. laevis* vocal circuit



Nucleus IX.X (control)

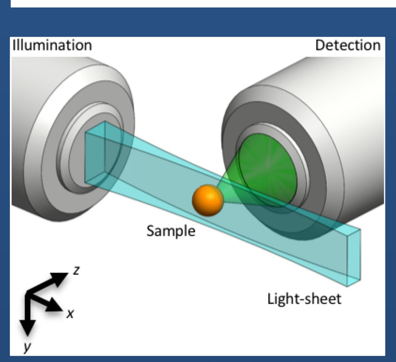
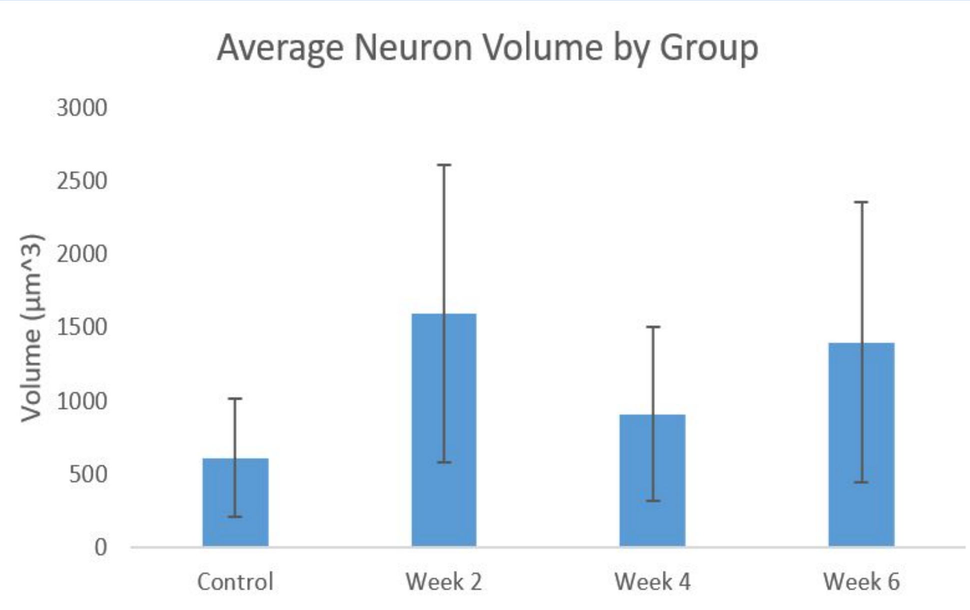


Diagram of lightsheet microscopy process

Cleared *X. laevis* brain

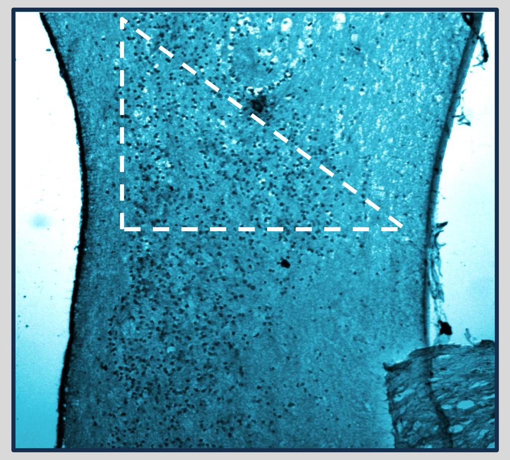
Nucleus IX.X (four week)



Preliminary comparison of average group soma size. Data obtained from discernible PBx neurons in viable images using Imaris software

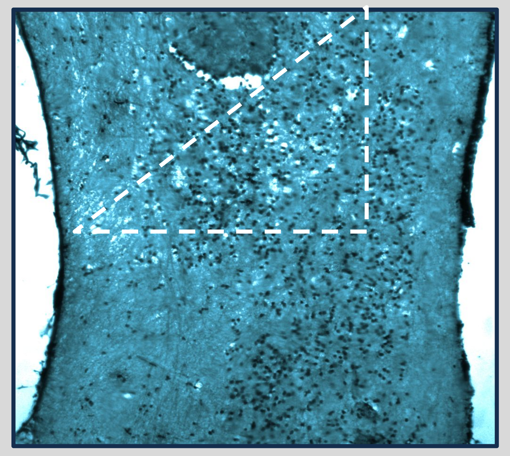
Are Target Vocal Neurons New or Pre-existing?

- Bromodeoxyuridine (BrdU) is a thymidine analog that marks cellular proliferation by incorporation into the DNA of S-phase cells.[1]
- We used BrdU IHC to attempt to mark possible testosterone induced neurogenesis in PBx.



PBx section of T-treated experimental female X. laevis

- We didn't find evidence of neurogenesis above levels of background staining.
- Could indicate that the target neurons are pre-existing.
- Or that there was too much background staining/one of the protocol steps failed.



PBx section of untreated control female X. laevis

Conclusion

- Results are inconclusive as to a timeline of PBx neuron growth between groups.
- This was largely due to issues with tissue clearing, loss of fluorescence, and time limitations.
- However, this project has resulted in more refined methods for testosterone administration, preparation of brains for imaging, and BrdU immunostaining.
- These developments will open up further exploration into how testosterone guides neuronal changes underlying vocal development.

Future Directions:

- **Develop methods of data acquisition and analysis**
- **Further optimize procedures for clearing fluorescently labeled frog brains**
- **Continue this study with a larger sample size and more streamlined, uniform methodology.**

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References

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